



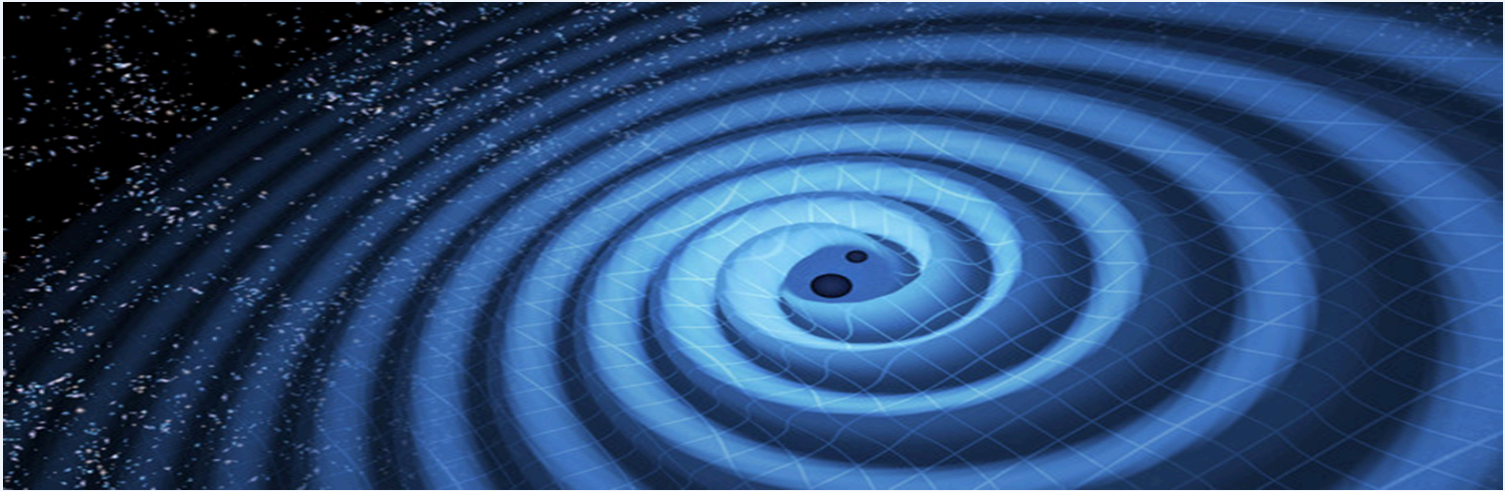
Gravitational Wave Detection

Summer Project Report

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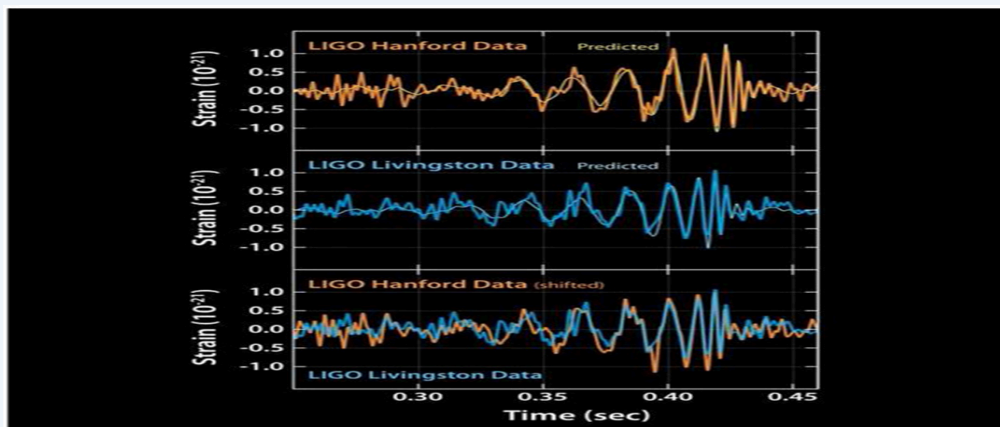
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Introduction & Theoretical Background

Gravitational waves are 'ripples' in space-time produced by massive accelerating objects (things like neutron stars or black holes orbiting each other). These waves travel with the speed of light, carrying with them information about their origins, as well as clues to the nature of gravity itself.

Gravitational wave detection requires sophisticated signal processing to identify weak astrophysical signals (they stretch and squeeze space by one part in 10^{21}) buried in environmental noise.



Sources and Types of Gravitational Waves

Astronomers have defined four categories of gravitational waves based on what object or system generates the waves:

- Compact Binary Inspiral Gravitational Waves
- Continuous Gravitational Waves
- Stochastic Gravitational Waves
- Burst Gravitational Waves

Compact Binary Inspiral Gravitational Waves:

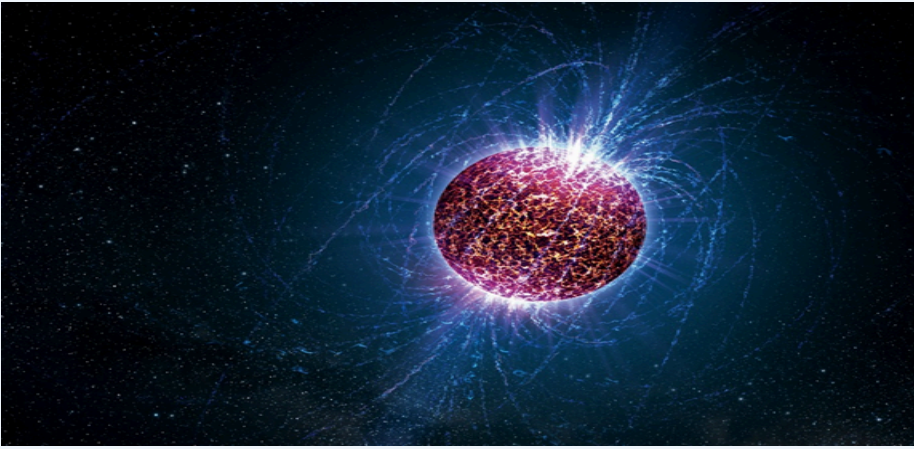
These waves are produced by orbiting pairs of massive and dense(“compact”) objects like blackholes and neutron stars. There are three subclasses of ‘compact binary’ systems in this category:



- 1.Binary Neutron Stars (BNS): two neutron stars orbiting each other.
- 2.Binary Black Hole (BBH): two black holes orbiting each other.
- 3.Neutron Star-Black Hole Binary (NSBH): a neutron star and a black hole orbiting each other.

Continuous Gravitational Waves:

These waves are expected to be produced by a single spinning massive object like a neutron star. Any bumps on or imperfections in the spherical shape of this star will generate gravitational waves as it spins.



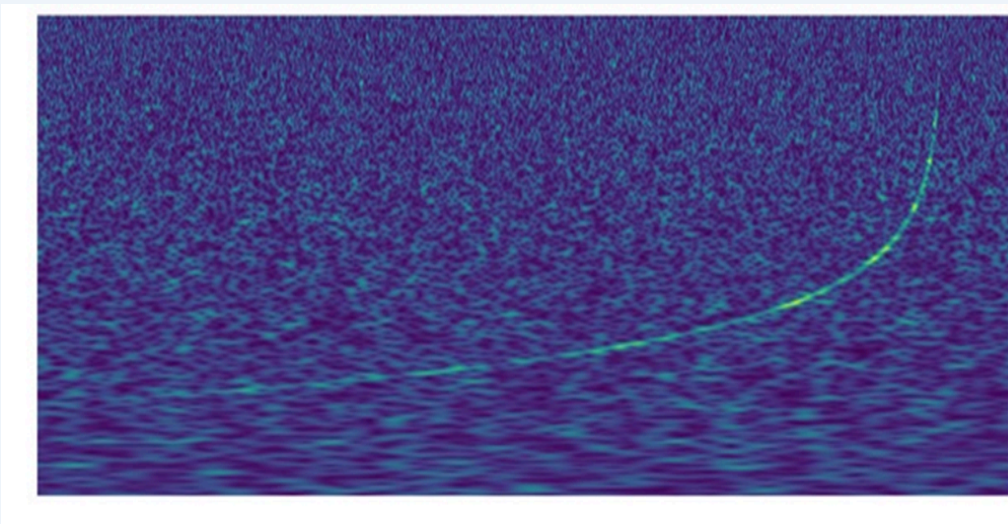
Stochastic Gravitational Waves:

These are small waves from every direction that make up what is called “Stochastic Signal”, so called because the word ‘stochastic’ means having a random pattern that may be analyzed statistically but not predicted precisely. These will be the smallest and most difficult gravitational waves to detect, but it is possible that at least part of this stochastic signal may originate from the Big Bang.

Burst Gravitational Waves:

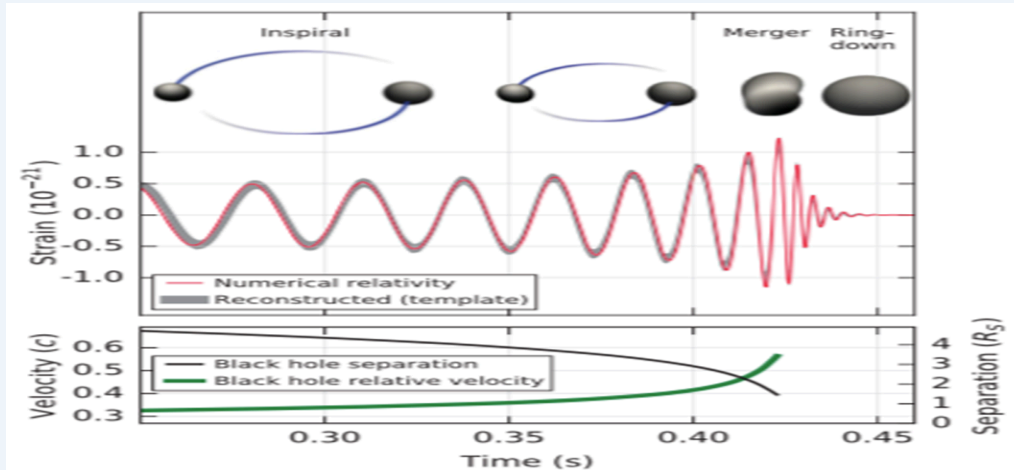
These waves come from short-duration unknown or unanticipated sources.

Chirps



It is the characteristic strain signal generated by the inspiral, merger, and ringdown of compact binary systems. The morphology is defined by a distinct waveform that continuously increases in both frequency and amplitude over time until a sudden peak, followed by a rapid decay.

The morphology is broken down into three distinct, mathematically modeled phases:



The Inspiral Phase

As two compact objects orbit one another, they lose energy by radiating gravitational waves. This causes their orbits to shrink, which forces the objects to rotate faster. This creates a slowly oscillating wave that gradually increases in frequency and amplitude. The rate at which frequency sweeps upward is dictated by the system's chirp mass.

The Merger Phase

The two bodies reach their closest stable orbit and plunge into each other to form a single, highly distorted object. This represents the cataclysmic climax of the chirp. The frequency and amplitude peak sharply and then reach an abrupt cutoff (often referred to as the 'trough' of the chirp) as the horizon settles.

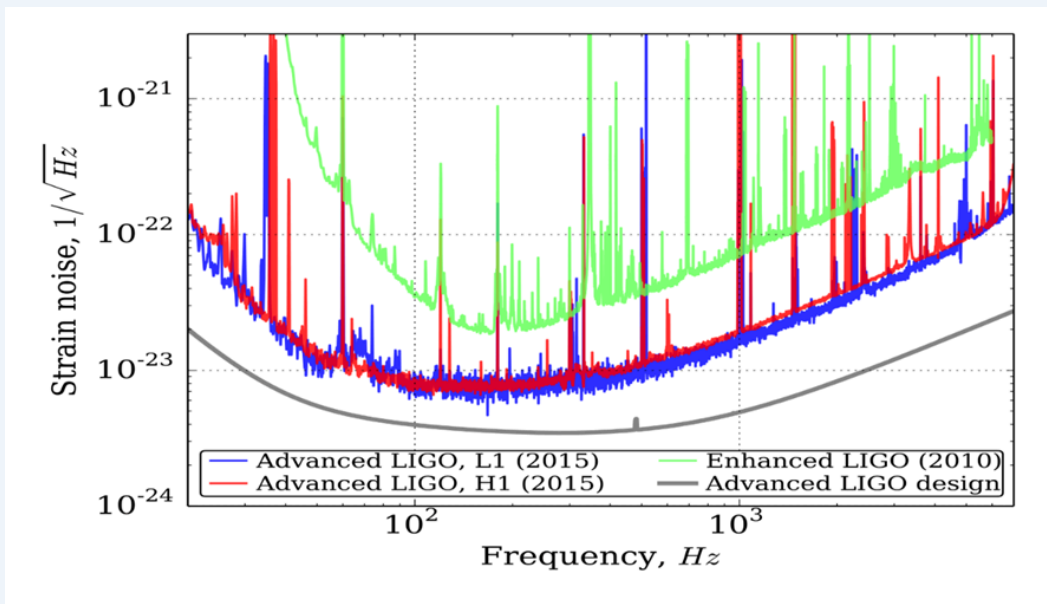
The Ringdown Phase

The newly formed, single black hole is initially highly deformed. To achieve a stable, symmetrical state, it vibrates and sheds its asymmetry by radiating gravitational waves. The signal rapidly decays in a series of damped, exponentially fading oscillations called quasi-normal mode.

Data Architecture & Sources

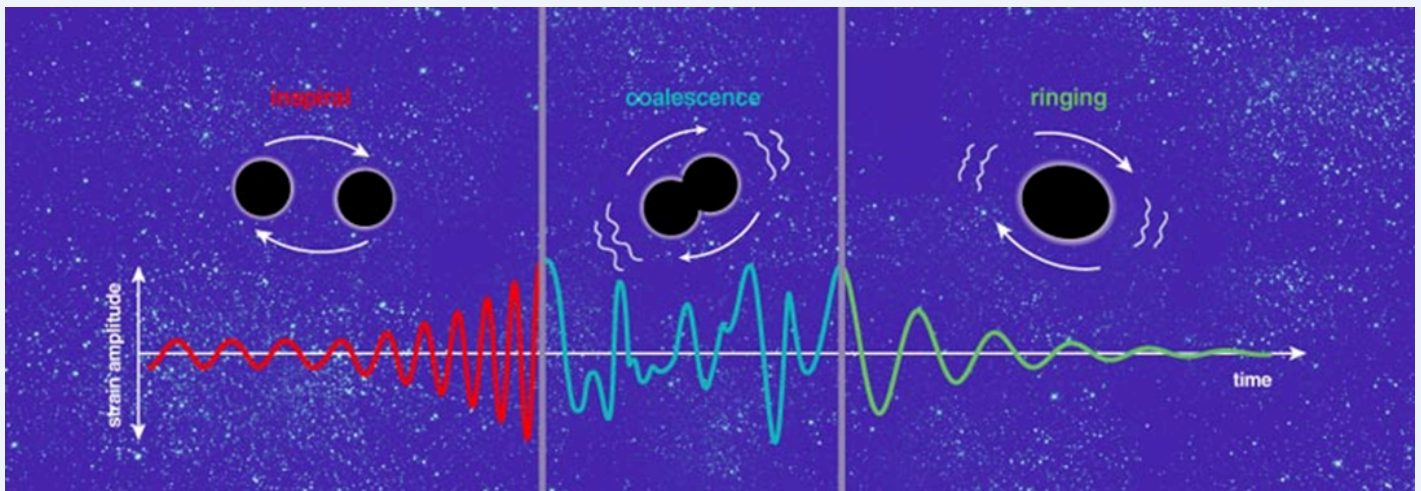
The GWOSC Ecosystem

The Gravitational-Wave Open Science Center (GWOSC) fulfills U.S. NSF LIGO’s commitment to release, archive and serve gravitational wave data to the broader scientific community and to use the public, and to provide the information and tools necessary to understand and use the data. The GWOSC site includes detailed metadata, tutorials, tools, and software to help public users perform effective analyses using NSF LIGO, Virgo, KAGRA data.



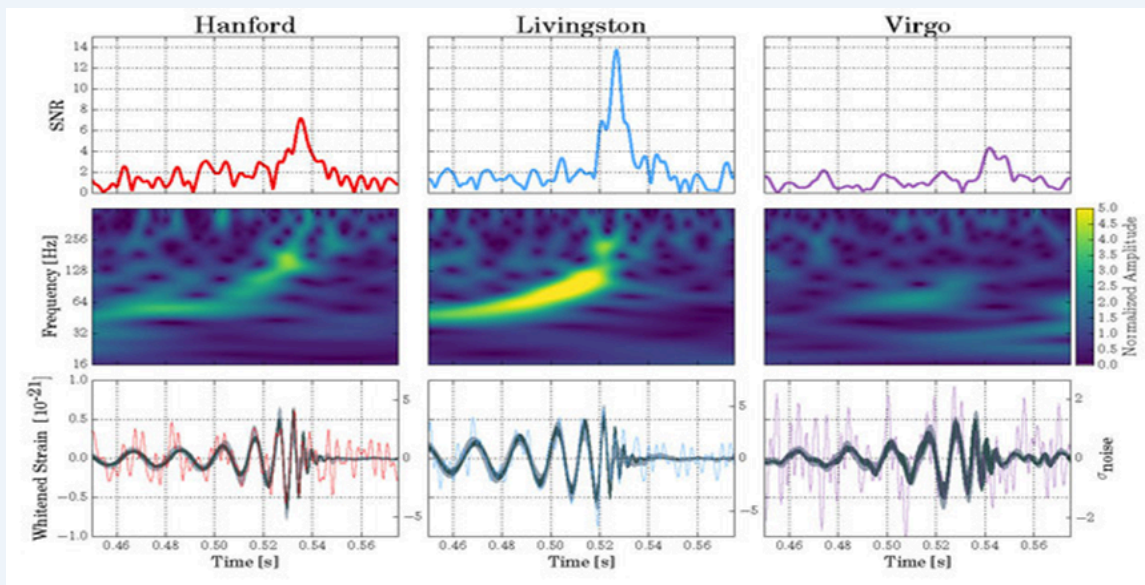
Metric	Bulk Data Releases	Event Portal Releases
Primary Purpose	Continuous background searches, continuous waves, stochastic background, population	Individual event characterisation waveform matching, parameter estimation .

	studies.	
Total Available Span	Months to Years	Focused window around a single timestamp
Discrete File Options	Choice of 4096 s or 32s chunks	Choice of 32 s, 128 s, 512 s, 4096 s
Data Continuity	Subject to gaps	Guaranteed continuity over the short segment
Available Timeline	Post-proprietary period	Released dynamically alongside official catalogue publication



The GWOSC structures its public data into two primary architectures: Bulk Data Releases and Event Portal Releases.

Data Simulation



The machine-learning stage requires many labelled examples, while confirmed gravitational-wave events are comparatively limited. For this project, the dataset is constructed by injecting simulated binary-black-hole (BBH) waveforms into real detector noise. The raw dataset is generated first; whitening, normalization and time-frequency transformations are applied later as part of the model preprocessing pipeline.

Dataset Objective

The main objective was to create a labelled dataset for **binary gravitational-wave detection**, where **label 1 represents GW signal + detector noise** and **label 0 represents noise only**. Simulated BBH waveforms were combined with real detector noise so that a sufficiently large and controlled dataset could be created for deep-learning training.

Signal Construction

The gravitational-wave signals were generated using **PyCBC** with the **IMRPhenomD** waveform model for binary black-hole mergers. A bank of **80 different waveform templates** was generated by varying the component masses from **20 to 75 solar masses**, with the mass

ratio restricted to the selected limit. Spin and eccentricity were not varied because the present task focuses on **signal detection rather than parameter estimation**.

The lower frequency of each waveform was adjusted when required so that its important merger-containing part could fit within the 1-second analysis window without severe truncation.

Signal Injection and SNR

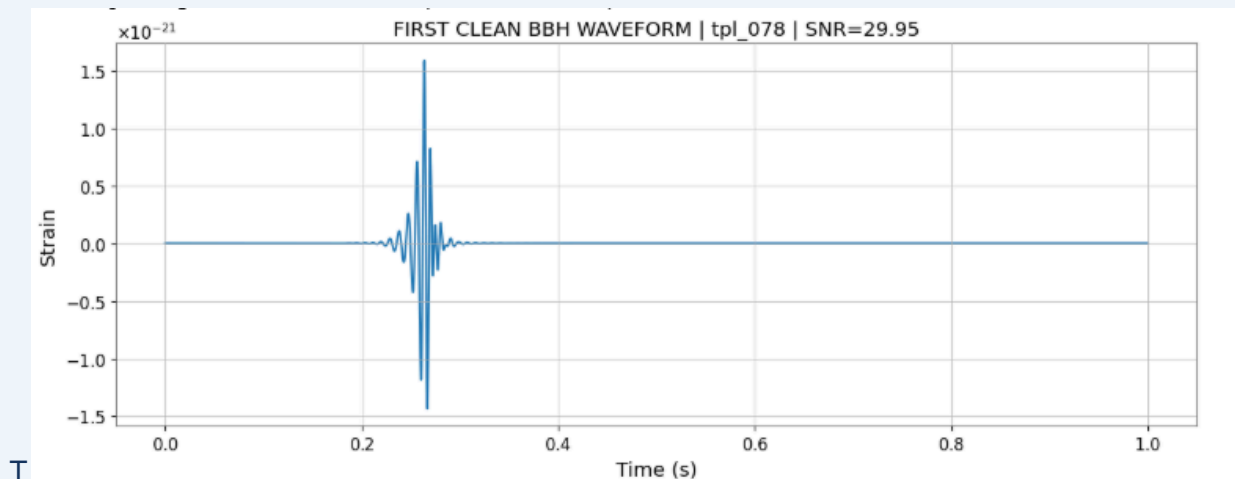
For each positive sample, a simulated gravitational-wave waveform was placed at a valid position inside a real 1-second detector-noise segment. Its amplitude was then scaled to a selected target **signal-to-noise ratio (SNR)** and added to the noise.

The target SNR was sampled mainly from the ranges **8–14, 14–20, and 20–30**, with the largest proportion of samples generated in the lower SNR range. This introduces signals with different strengths and also includes more challenging low-SNR examples.

Thus, the positive class can be represented as:

GW + Noise=Simulated BBH waveform+Real H1 detector noise.

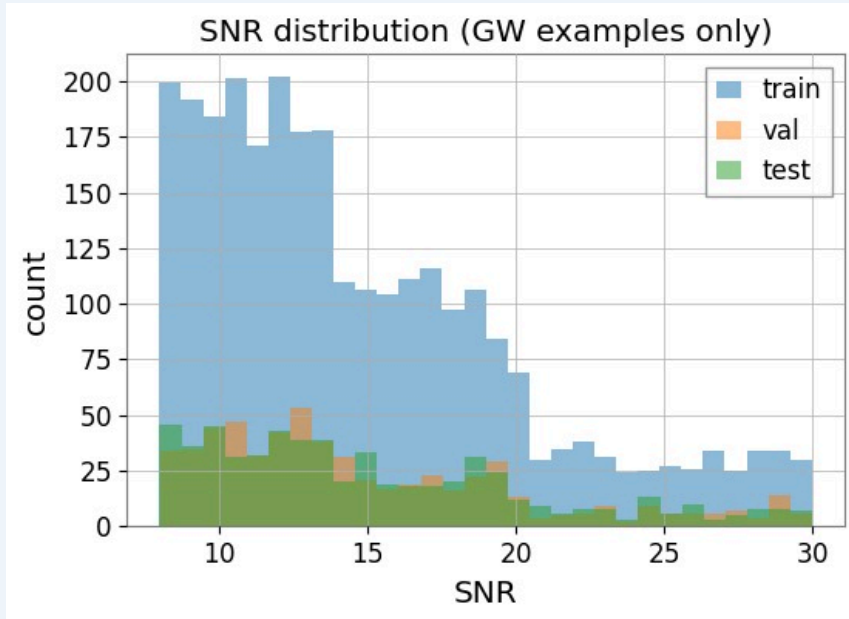
The negative class consists only of real detector-noise segments.



Template-Disjoint Dataset Split

An important part of the dataset design is that the waveform templates were **completely separated between training, validation, and testing**. The 80 templates were divided into **56 training templates, 12 validation templates, and 12 testing templates**, and no template ID is shared between the three sets.

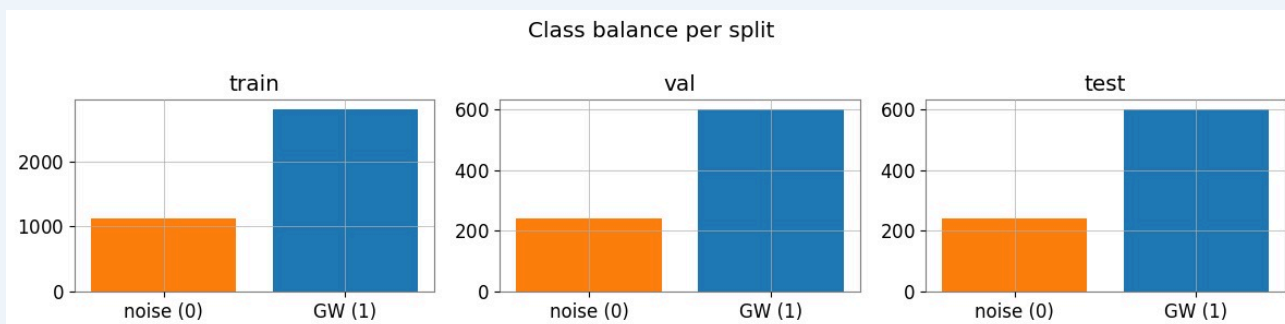
This prevents the same waveform morphology from appearing in both training and testing. Although multiple samples can be generated from one template using different noise segments, SNRs, and positions, that template is restricted to only one split. Therefore, the test set evaluates the model on **unseen waveform templates**, reducing template-level data leakage.



Dataset Structure and Storage

Using 50 GW + noise examples for each template produced **2800 training, 600 validation, and 600 testing positive samples**. A total of **1600 noise-only samples** were distributed across the three splits. The final dataset therefore contained **3920 training samples, 840 validation samples, and 840 testing samples**.

The dataset was stored in **HDF5 format** along with metadata such as the label, template ID, component masses, SNR, noise-segment ID, split, and waveform peak position. The generated strain was kept **unwhitened at this stage**. Whitening and spectrogram generation were performed later as separate preprocessing steps for the 1D-CNN and pretrained ResNet-18 branches.



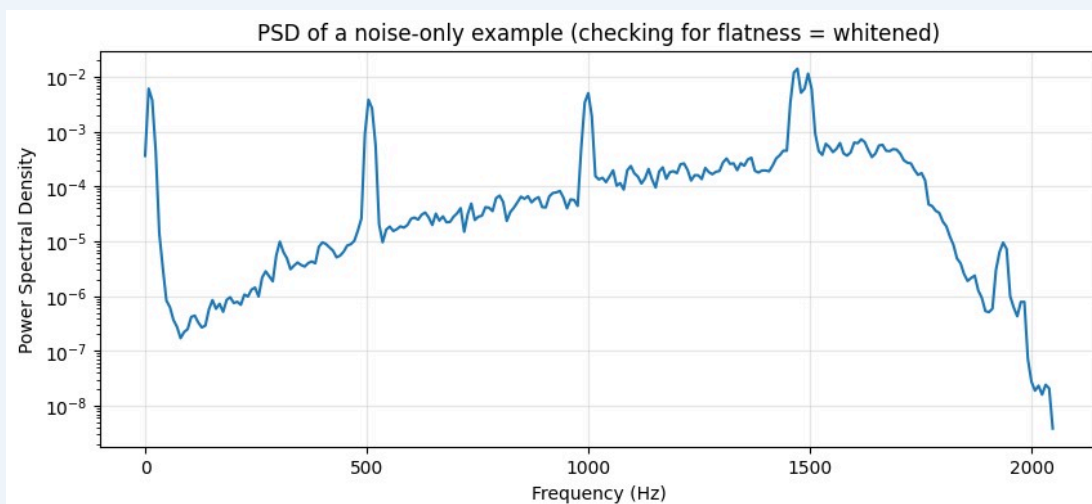
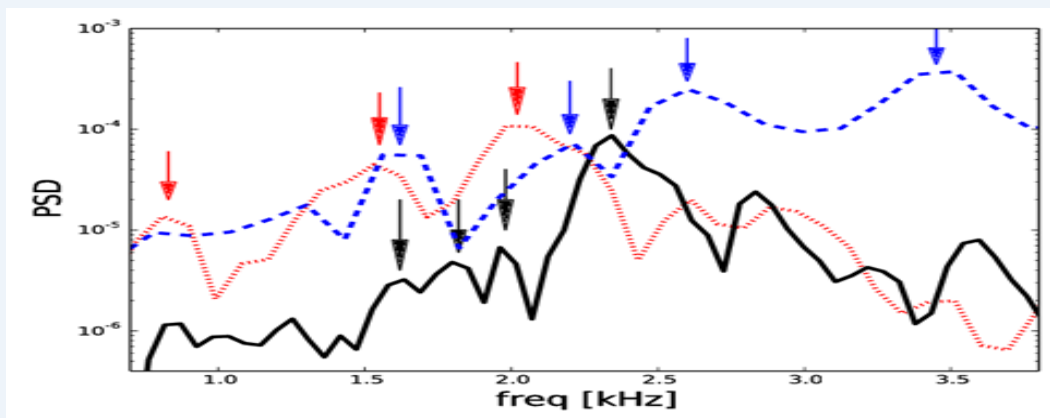
Digital Signal Processing & Preprocessing

Power Spectral Density

When gravitational waves travel through space they stretch and squeeze distances. This effect is measured as strain (h), which is a dimensionless number. Strain tells us the change in length with respect to the original length.

$$h = \Delta L / L$$

Power Spectral Density (PSD), $S_h(f)$, is a statistical tool used to analyze how much power a random, fluctuating signal holds at different frequencies.



Characteristic Strain(h_c)

While PSD is mathematically foundational for raw data analysis, its physical unit (Hz^{-1}) makes it difficult to visually interpret signals directly from a plot. Characteristic strain is a dimensionless quantity that scales the square root of the power spectral density by the frequency at which it is measured.

Signal to Noise Ratio

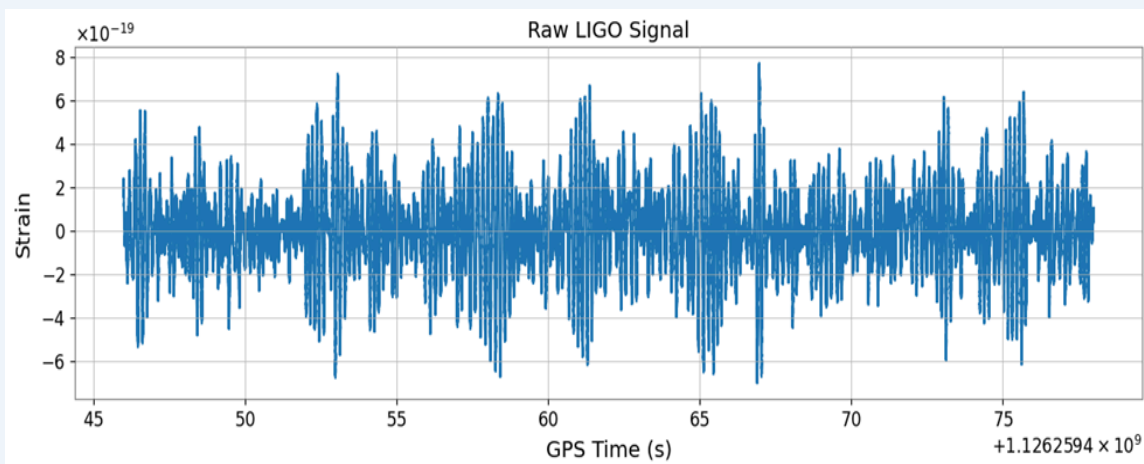
Since gravitational wave signals are often weak and buried deep within complex instrument noise, simple threshold detection is insufficient. Instead, methods like matched filtering are used.

Precise computer models, called templates, simulate what a gravitational wave should look like when black holes or neutron stars collide. The computer then glides these templates across the noisy detector data, looking for a perfect match. To determine if a template actually matches the data, the computer calculates a score called Signal-to-Noise Ratio (SNR), represented by ρ .

The matched filter tracks the ‘chirp’ across the detector’s quietest frequency ranges, continuously accumulating statistical confidence over the duration of the event. If the template lines up perfectly with a real wave hidden in the data, the SNR score will experience a massive, unmistakable spike.

Signal Processing Pipeline

Raw Signal

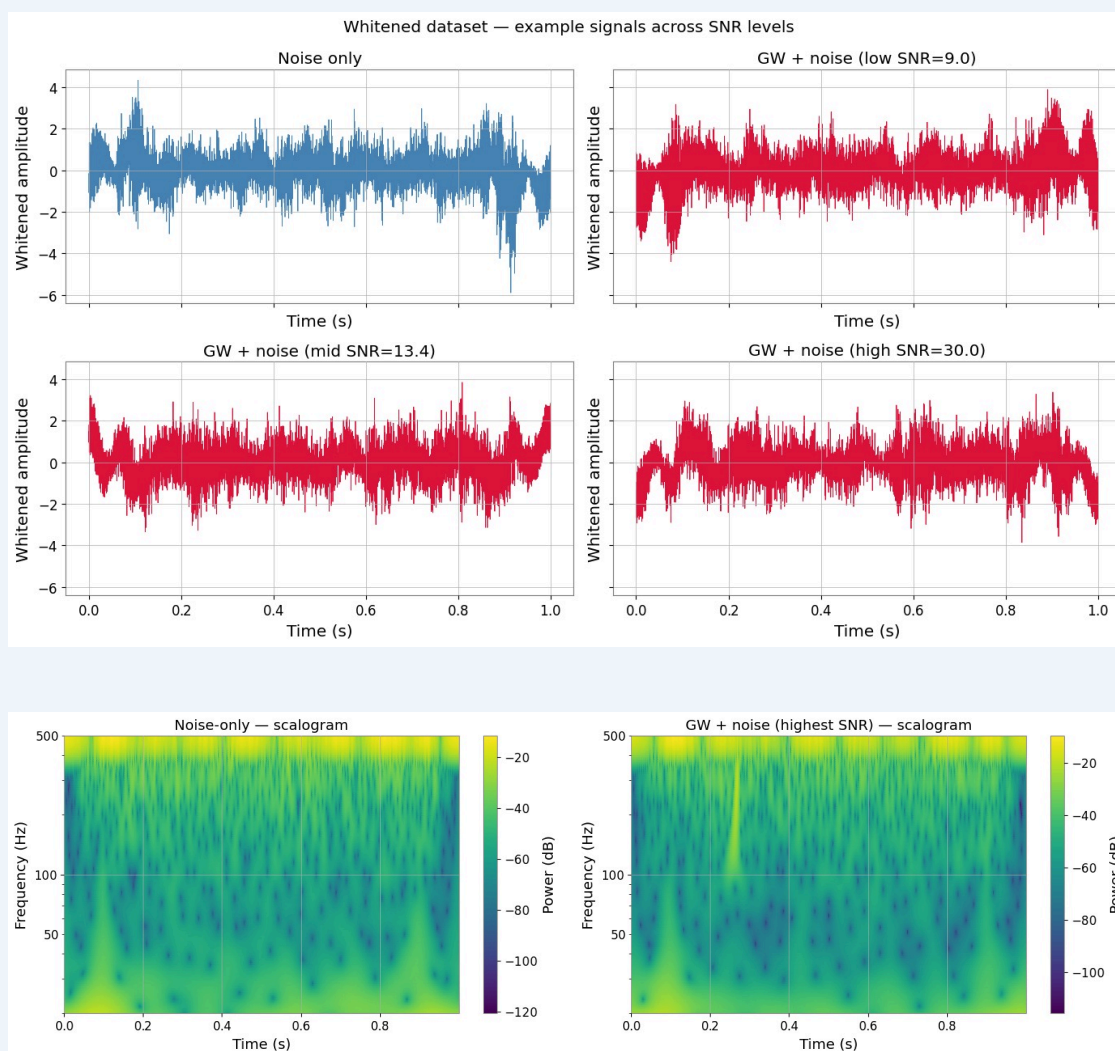


Whitening

It is a frequency-domain operation that normalizes data so that its background noise power is completely uniform across all frequencies.

White noise contains all frequencies of sound or data at equal power.

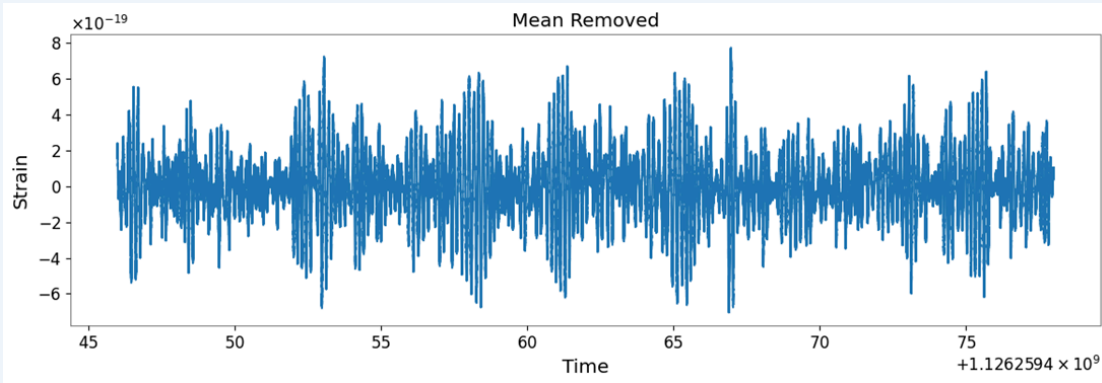
Gravitational wave detectors suffer from colored noise. The detector's sensitivity fluctuates wildly across the frequency spectrum. If one tries to search for a gravitational wave in raw data, the loud frequencies will completely dominate mathematical algorithms, while the quiet frequencies will be completely ignored.



Mean Removal (Zero-Centering):

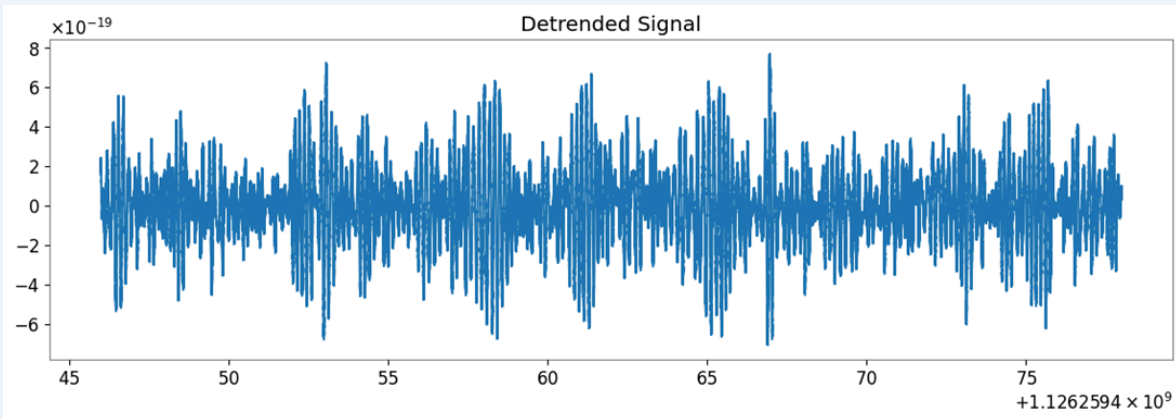
The mathematical average(mean) of the entire data segment is calculated and subtracted from every single data point in that segment. It forces the time-series to fluctuate exactly around zero.

$$h_{\text{conditioned}}(t) = h_{\text{raw}}(t) - \langle h_{\text{raw}}(t) \rangle$$



Detrending

It addresses non-constant, slow macroscopic drifts across the data segment. This process flattens the baseline of the data.



Bandpass Filtering

It is a filtering technique that isolates a specific window of frequencies while blocking everything outside of it. A bandpass filter relies on two distinct frequency thresholds, known as cutoff frequencies-

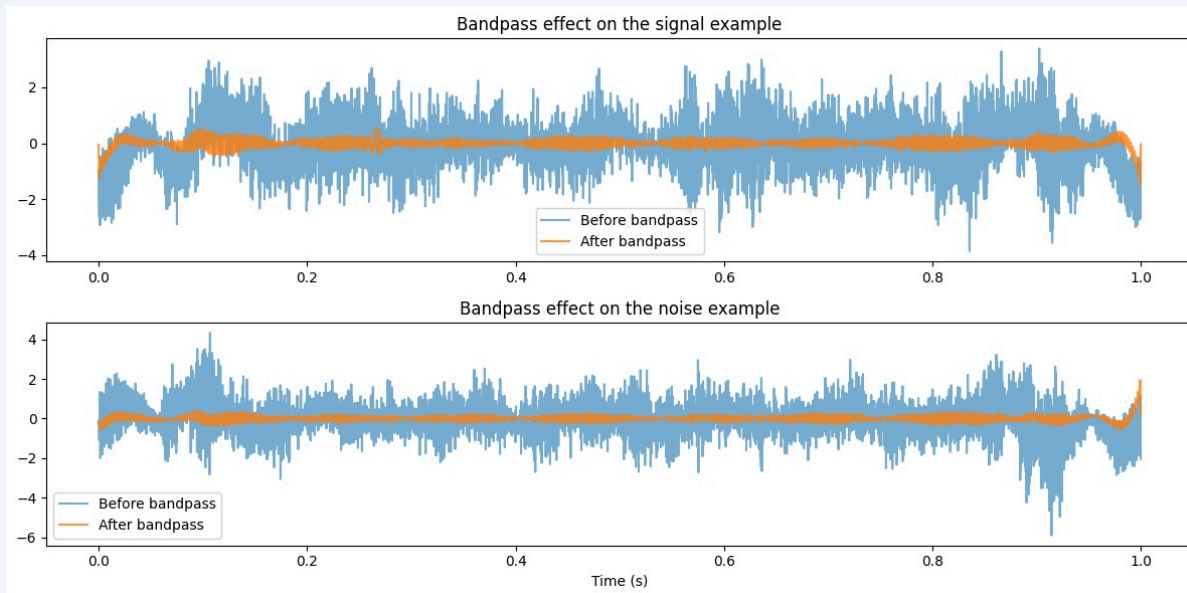
- **The Lower Cutoff(f_{low}):**

Blocks frequency below this threshold. This functions as a high pass filter to eliminate seismic rumble, ocean waves, and suspension thermal noise (typically below 20-40 Hz).

• The Upper Cutoff(f_{high}):

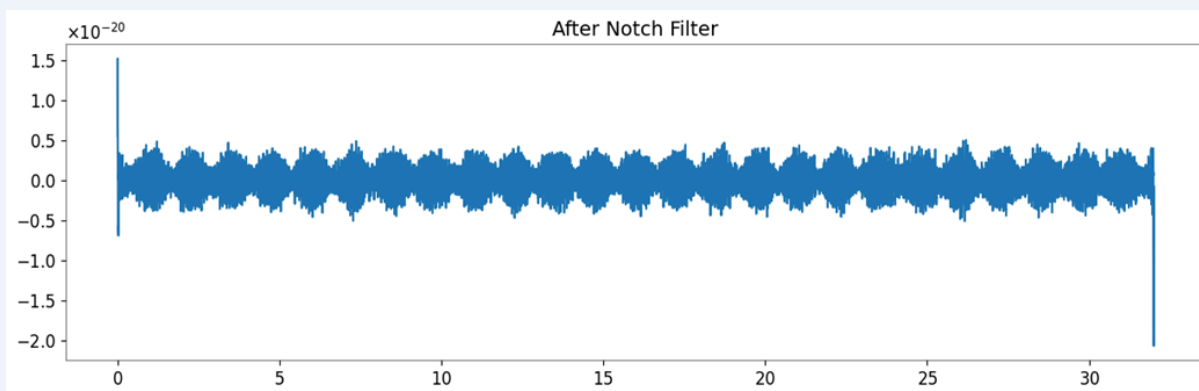
Blocks frequencies above this threshold. This functions as a low-pass filter to eliminate high frequency photon shot noise and digital quantization noise (typically above 300-500 Hz).

The space between these two thresholds is called the passband. Any data falling inside this passband is allowed through completely uninhibited.

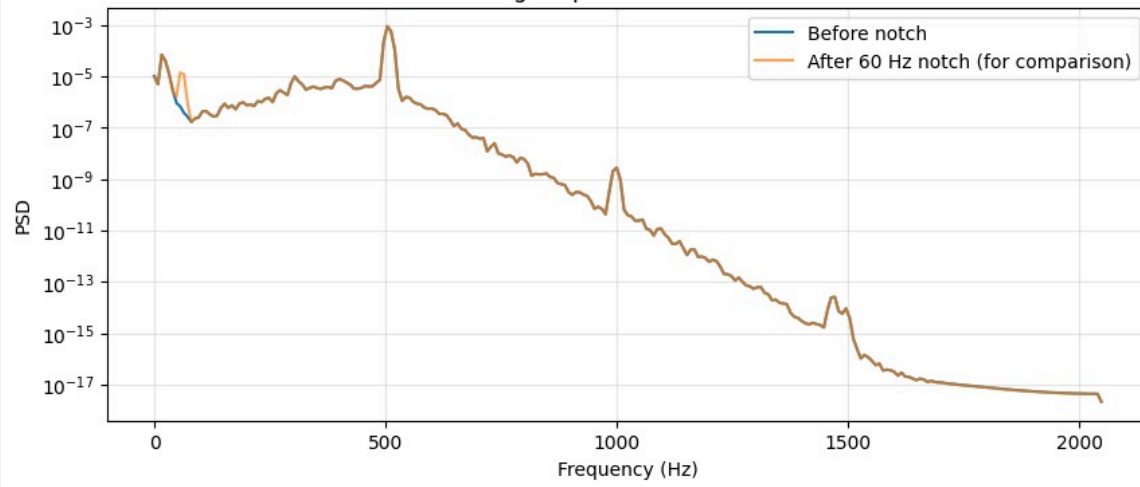


Notching

A notch filter is a specialized signal processing filter that attenuates a highly specific, ultra-narrow slice of frequencies while leaving everything untouched.



Checking for power-line interference



Time Frequency Representation: Spectrogram

Fast Fourier Transform (FFT)

Gravitational wave data streams are continuous, high frequency time series (typically sampled at 4096 Hz or 16384 Hz). The Fast Fourier Transform (FFT) converts recorded detector strain data from the time domain onto the frequency domain, which makes detecting weak gravitational wave signals computationally possible.

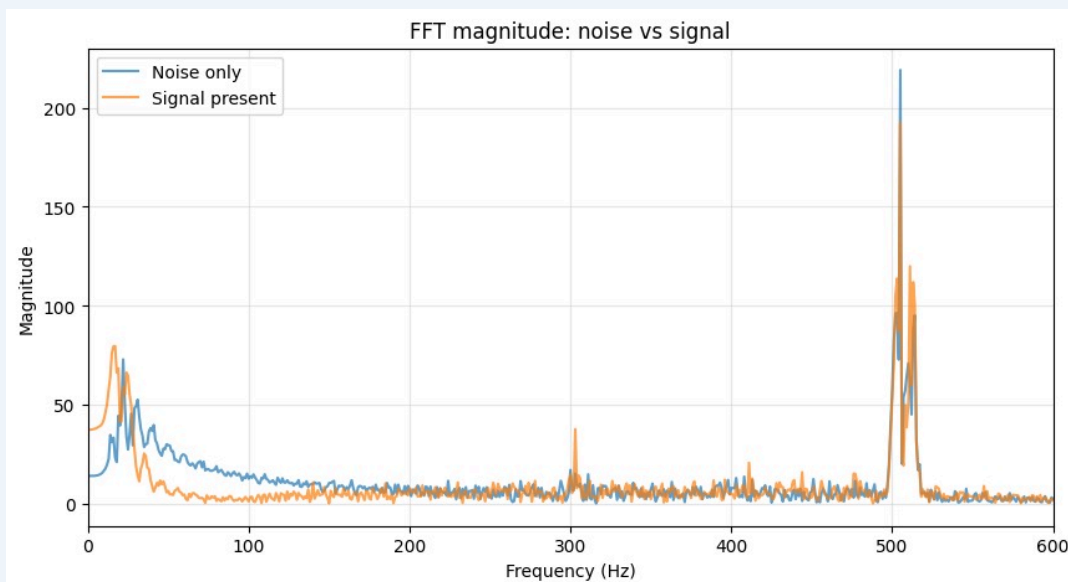
Applying FFT enables point-by-point algebraic operations to replace complex time-domain convolutions.

Matched Filtering:

Detectors like LIGO compare noisy data against theoretical waveform templates. Doing this cross-correlation directly in the time domain is too slow. Using the FFT turns slow convolution into fast point by point multiplication in the frequency domain.

Noise Whitening and Power Spectral Density:

Gravitational wave data is full of colored detector noise. FFT may be used to estimate the noise PSD and “whiten” the data, turning the noise into an uncorrelated, flat baseline so hidden signals stand out.



Short-Time Fourier Transform(STFT)

Short-Time Fourier Transform maps how the signal's power spectral density evolves in both time and frequency simultaneously.

STFT breaks down a continuous, long time-series strain data stream $h(t)$ into small, overlapping time segments using a localized sliding window $w(t)$. The Fourier Transform is then computed for each individual segment.

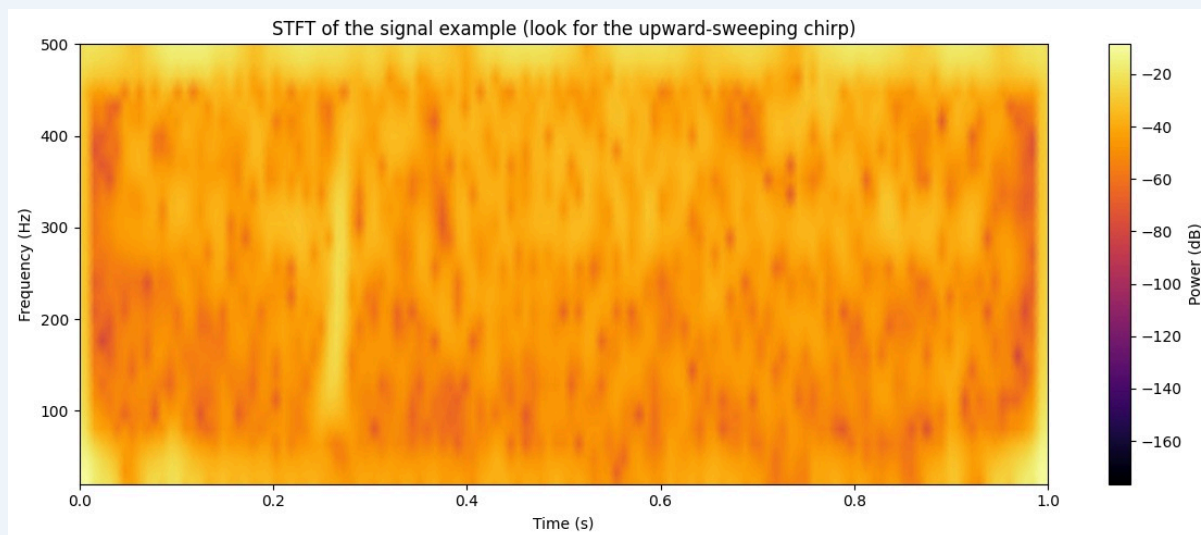
Squaring the magnitude of the STFT yields the spectrogram, which provides 3D visualization (Time vs. Frequency vs. Power) of the gravitational wave energy.

Noise Handling:

Efficiently manages non-stationary detector noise and sudden data gaps.

Modulation Tracking:

Accommodates detector-induced signal modulations caused by Earth's rotation or orbital motion.



Continuous Wavelet Transform

The Continuous wavelet transform maps localized strain data onto a simultaneous timefrequency architecture. The CWT scales and translates a localized, finite multi-resolution mother wavelet function.

Glitch Detection and Vetoing:

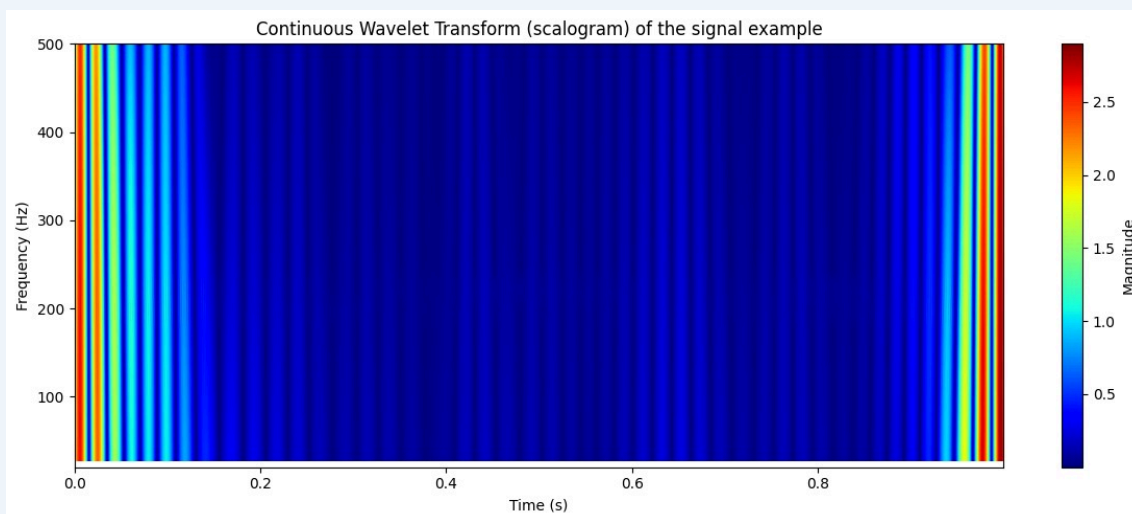
Interferometric detectors are prone to glitches. Standard FFT globally spreads these signals, corrupting vast blocks of data. Time-Frequency wavelet pipelines isolate glitches as localized excess-energy patches on a multiresolution space. This allows data processing pipelines to automatically flag veto, or perform coherent subtraction of transients without losing surrounding signal integrity.

Unmodeled Transient (Burst) Searches:

While matched filtering relies on precise theoretical templates (e.g., binary black hole mergers), some cosmic events lack exact models—such as core-collapse supernovae or cosmic string loops. Search algorithms like Coherent Wave-Burst (cWB) use a Wilson-Daubechies time-frequency wavelet transform to search for localized excess energy across multiple detectors simultaneously without needing a prior waveform template.

Tracking the "Chirp" Profile:

As compact binaries inspire and merge, the emitted wave continuously sweeps upwards in both frequency and amplitude over time (a chirp). The CWT perfectly adapts to this geometry: it provides the high time resolution required to capture the rapid, late-stage merger phase and the high frequency resolution needed to track the slow, early inspiral phase.



Machine Learning Pipeline

After generating the gravitational-wave dataset, two different deep learning approaches were used for binary classification of **gravitational-wave signals and detector noise**. The main purpose of using two models was not simply to compare their accuracy, but to test two different representations of the same gravitational-wave data.

The first model was a **custom 1D Convolutional Neural Network (1D CNN)** trained directly on the **whitened raw strain time series**. In this approach, the network received the gravitational-wave strain as a one-dimensional signal without converting it into an image.

The second approach used the **signal-processed spectrograms** generated from the dataset. These spectrograms represent the data in the time-frequency domain and were given to an **ImageNet-pretrained ResNet-18 model**. Instead of training ResNet from random initialization, the pretrained ImageNet weights were used as the starting point and the final classification layer was replaced for the two-class gravitational-wave detection problem.

Model A: 1D CNN on Whitened Raw Strain

Data

It worked directly with the **whitened gravitational-wave strain**. Each sample represented a one-second segment sampled at 4096 Hz, giving 4096 strain values for one input example.

Therefore, the original input shape was:[4096]

and it was converted to: [(1,4096)]

before being passed to the convolutional network. The additional dimension represents the single input channel required by the **Conv1d** layer.

CNN Architecture

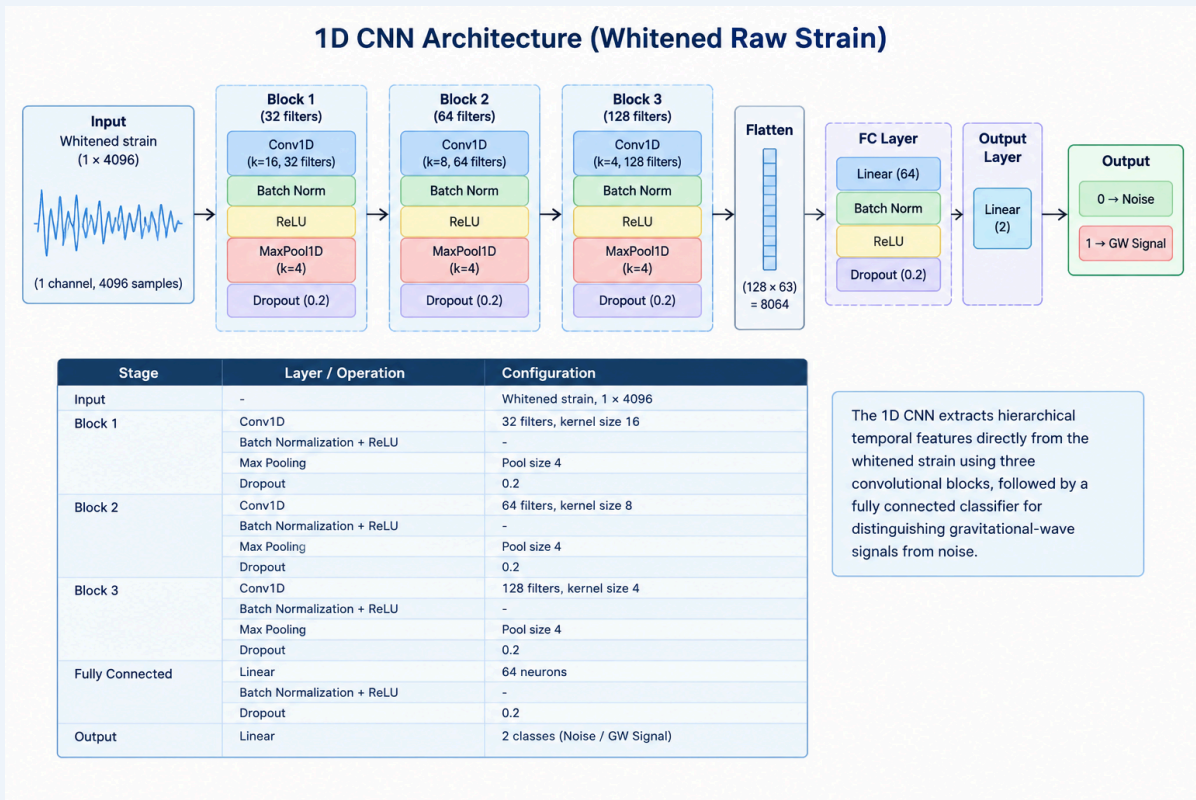
A three-block 1D CNN was used for this task.
The architecture was repeated for three convolutional blocks.

Conv1D ⇒ Batch Normalization ⇒ ReLU ⇒ Max Pooling ⇒ Dropout

After extracting features from the strain signal, the feature maps were flattened and passed through a fully connected layer containing 64 neurons. A final linear layer produced two outputs corresponding to:

[0={Noise}] and [1={Gravitational Wave}]

A dropout value of 0.2 was used to reduce unnecessary dependence on individual features.



Training

The training data was shuffled within each split while keeping the training, validation, and test sets separate. Since the training set was

imbalanced, a `WeightedRandomSampler` was used so that the minority noise class was sampled more frequently during training. This helped prevent the model from simply learning the majority-class distribution.

The model was trained using the **Adam optimizer** with a learning rate of (10^{-3}) . Cross-entropy loss was used as the classification loss. Gradient clipping with a maximum norm of 1.0 was also applied during backpropagation.

A `ReduceLROnPlateau` scheduler was used to reduce the learning rate when validation AUC stopped improving. Early stopping was also implemented, with validation AUC used as the main metric for deciding whether the model was improving.

This choice was important because the dataset was class-imbalanced. Accuracy alone could give a misleading result if the model became biased toward one class. AUC was therefore used to monitor the actual separation between the gravitational-wave and noise classes.

Evaluation Metrics

Several evaluation metrics were calculated after each validation epoch.

Accuracy measures the fraction of correctly classified samples:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision measures how many samples predicted as gravitational-wave signals were actually signals:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall measures how many of the actual gravitational-wave signals were successfully detected:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1-score combines precision and recall:

$$\text{F1} = 2 \cdot [(\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})]$$

The **ROC-AUC** was used as the main model-selection metric. A value near 1 indicates strong class separation, whereas a value near 0.5 indicates performance close to random discrimination.

The confusion matrix was also used during final testing to examine the numbers of true negatives, false positives, false negatives, and true positives separately.

1D CNN Training Results

The 1D CNN showed increasing training accuracy, but this improvement did not translate to better validation performance. Training accuracy increased from **0.516 in epoch 1 to approximately 0.649 by epoch 21**, while validation AUC peaked at **0.550 in epoch 1** and remained close to **0.5** thereafter.

Although validation accuracy and F1-score increased at some later epochs—for example, **0.710 accuracy, 0.970 recall, and 0.827 F1-score at epoch 11**—the corresponding AUC was only **0.484**. This indicates that the model could achieve high threshold-based metrics by predicting many samples as signals, but its overall ability to distinguish signals from noise remained weak.

Therefore, **validation AUC was used for model selection**, with the best checkpoint occurring at **epoch 1 (AUC = 0.550)**. Training stopped at epoch 21 after the validation AUC failed to improve for the configured patience period.

Results

The final test results were:

Metric	Result
Test Loss	0.7306
Test AUC	0.538
Test Accuracy	0.324
Test F1-score	0.142
Test Precision	0.758
Test Recall	0.078

confusion matrix was:

225	15
553	47

where the rows correspond to the actual class and the columns correspond to the predicted class.

Interpretation

The 1D CNN showed limited generalization on the unseen data. Although the training accuracy gradually increased, the validation AUC remained close to 0.5 for most of the training, indicating weak separation between GW signals and noise.

The final test AUC was **0.538**, while the recall was only **0.078**. The confusion matrix showed that **553 out of 600 GW signals were classified as noise**, meaning that the model missed a large number of actual signals. Although the precision was 0.758, the low recall resulted in a low F1-score of 0.142.

Overall, the results indicate that the current 1D CNN was able to learn from the training data but did not learn features that generalized well to unseen raw strain samples.

Possible Improvements for the 1D CNN

The performance of the 1D CNN could be improved by increasing the **diversity of the training dataset**, particularly by using a wider range of waveform parameters, signal-to-noise ratios, noise segments, and injection positions. This can help the model learn general gravitational-wave characteristics instead of dataset-specific patterns.

The CNN architecture and training settings could also be optimized. Experiments with a **smaller learning rate, weight decay, different dropout rates, and a more compact architecture** may improve generalization. Global average pooling could also be considered to reduce the number of parameters before the fully connected layers.

Further improvement could come from better **input scaling and normalization** after whitening, along with suitable data augmentation for the raw strain signal. Finally, validation AUC, recall, and F1-score should continue to be monitored during training so that the model is selected based on its ability to detect unseen GW signals rather than training accuracy alone.

Model B: ImageNet-Pretrained ResNet-18 on Spectrograms

Spectrogram Representation

The second model used **signal-processed spectrograms** rather than directly using the one-dimensional strain. A spectrogram represents the signal in both time and frequency, allowing the model to observe how frequency content changes with time.

The generated spectrograms had the shape: $129 * 125$

Before entering the neural network, the spectrogram values were normalized using the mean and standard deviation calculated only from the training set. The calculated values were:

mean=-8.5209

and

standard deviation=3.3551

The same training-set statistics were then applied to the validation and test sets. This prevented information from the validation or test datasets from influencing the preprocessing stage.

Pretrained ResNet-18

Instead of training a ResNet-18 network from scratch, an **ImageNet-pretrained ResNet-18** was used. The original ResNet-18 expects three-channel image input:

$[(N, 3, H, W)]$

whereas the generated spectrograms initially had only one channel:

$[(N, 1, H, W)]$

To make the spectrogram compatible with the pretrained architecture, the single channel was repeated three times. This allowed the original convolutional layers of ResNet-18 to remain unchanged while still accepting the spectrogram input.

The original ImageNet classification head was then replaced with:

Dropout(0.2)→Linear(512,2)

Thus, the model produced two output classes corresponding to noise and gravitational-wave signal.

The model was initialized using the default pretrained ResNet-18 weights. To account for this imbalance, class-weighted cross-entropy loss was used, with weights of 1.75 for noise and 0.70 for the GW class.

The network was then optimized for the gravitational-wave classification task using Adam. The learning rate was set to (10^{-4}) and the weight decay was (10^{-5}) .

ResNet Training using Validation AUC

The first ResNet experiment used validation AUC as the early-stopping and checkpoint-selection metric.

The model learned very quickly. In the first epoch, it achieved a training accuracy of 0.742 and validation AUC of 0.882. By epoch 2, validation AUC increased to 0.990. At epoch 3, it reached 0.9963, while training accuracy reached 0.984.

Although the training accuracy continued increasing in later epochs, the validation performance started fluctuating. For example, validation accuracy dropped strongly in epoch 4 even though training accuracy increased. At epoch 5, validation accuracy increased to 0.979 and validation AUC remained very high, but by epoch 6 validation accuracy dropped again to 0.750 and validation loss increased to 1.6651.

This behaviour shows that the pretrained ResNet was capable of learning the spectrogram patterns very quickly, but continuing to train did not consistently improve behavior.

The best validation AUC was 0.9963, obtained at epoch 3. Training stopped at epoch 6 because the validation AUC had not improved for the configured patience period. The best checkpoint was then restored before performing the final test evaluation.

Final Test Results of AUC-Monitored ResNet:

Metric	Result
Test Loss	0.6707
Test Accuracy	0.8524
Test AUC	0.9923
Test F1-score	0.8850
Test Precision	0.9979
Test Recall	0.7950

The confusion matrix was:

239	1
123	477

The test AUC of 0.9923 indicates that the model was able to separate the two classes very effectively. Its precision of 0.9979 means that almost every sample predicted as a gravitational-wave signal was actually a gravitational-wave sample. At the same time, the recall of 0.7950 shows that the model successfully detected a large majority of the gravitational-wave signals, although some signals were still missed.

ResNet Training Using Validation Accuracy

A second ResNet run was performed with validation accuracy as the model-selection and early-stopping metric. The architecture, dataset, normalization, learning rate, weight decay, and other major training settings remained the same. The only important difference was the metric used to decide which checkpoint was considered the best. The validation accuracy increased from 0.714 in the first epoch to 0.979 at epoch 5. The best validation accuracy was 0.9786 at epoch 5. However, after epoch 5, the validation accuracy fell sharply even though training accuracy continued increasing. Early stopping therefore occurred at epoch 8.

The final test results for the accuracy-monitored checkpoint were:

Metric	Result
Test Loss	1.7628
Test Accuracy	0.8060
Test AUC	0.9875
Test F1-score	0.8431
Test Precision	0.9977
Test Recall	0.7300

The confusion matrix was:

239	1
162	438

The model still performed well, but compared with the AUC-selected checkpoint, it missed more gravitational-wave signals: 162 instead of 123. Consequently, recall decreased from 0.7950 to 0.7300 and F1-score decreased from 0.8850 to 0.8431.

Overall Results and Discussion

The two models showed substantially different performance despite solving the same binary classification task. The **1D CNN**, trained on whitened raw strain, achieved a test AUC of only **0.538** and recall of **0.078**, indicating poor generalization and many missed gravitational-wave signals. Although training accuracy increased, validation AUC remained close to 0.5.

In contrast, the **pretrained ResNet-18**, trained on signal-processed spectrograms, achieved a test AUC of **0.9923**, accuracy of **0.8524**, F1-score of **0.8850**, precision of **0.9979**, and recall of **0.7950**. This suggests that the spectrogram representation provided a more informative feature space for the pretrained image-based CNN.

A likely reason is that spectrograms expose the signal’s time-frequency structure, allowing ResNet to learn spatial patterns effectively, whereas the smaller 1D CNN had to learn relevant waveform features directly from raw strain.

These results apply specifically to the generated dataset and current experimental setup, rather than proving that spectrogram-based models are universally superior. The ResNet also reached high validation AUC within a few epochs, making **epoch 3**, the best AUC checkpoint, the appropriate final model rather than the last training epoch.

Conclusion

- Overall, the experiment shows that the representation of gravitational-wave data has a major effect on the performance of a deep learning detector.
- The 1D CNN successfully completed the full training pipeline, but its final test performance showed weak generalization. The test AUC of 0.538 was close to random discrimination, and the recall of 0.078 showed that most gravitational-wave signals were missed.
- The signal-processed spectrogram combined with a pretrained ResNet-18 provided a much stronger and more generalizable classification result than the custom 1D CNN operating directly on whitened strain.
- The final pipeline therefore demonstrates that deep learning can be applied at two stages of the gravitational-wave signal-processing workflow: directly on the whitened time-domain signal and on its time-frequency representation.

This project investigated the use of digital signal processing and deep learning for the detection of gravitational-wave signals buried in detector noise. Two deep-learning approaches were evaluated for the same binary classification task. The comparison shows that converting the signal into a time-frequency representation exposed the characteristic structure of gravitational-wave events more effectively than directly processing the raw one-dimensional strain with the custom CNN. The strong performance of the pretrained ResNet-18 also demonstrates the potential benefit of transfer learning when working with limited scientific datasets.

Overall, the project demonstrates that gravitational-wave detection can benefit significantly from combining traditional signal-processing techniques with modern deep-learning methods. The experiment highlights a central lesson for AI-based scientific detection: model performance depends not only on the architecture but also on how the underlying physical data is represented. A robust future system could build upon this pipeline by improving dataset diversity, incorporating multiple detectors, testing additional time-frequency representations, and optimizing the detection model specifically for high recall while maintaining low false-alarm rates.